

Scattered C_0 tensor subcases for the CWO tensor question

Run `fa_banach_001`, lane 12

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Source and question

The source paper is

T. A. Abrahamsen, J. Becerra Guerrero, R. Haller, V. Lima, and M. Poldvere, *Banach spaces where convex combinations of relatively weakly open subsets of the unit ball are relatively weakly open*, arXiv:1806.10693.

In Section 6, the authors ask:

If both X and Y have property CWO, does the injective tensor product $X\widehat{\otimes}_\varepsilon Y$ also have property CWO? They also ask the analogous question for CWO-S and CWO-B.

b. QUESTIONS

We end the paper with some natural questions.

- (a) All of our infinite dimensional examples of spaces with property CWO contain c_0 , that is, both $C_0(K, X)$ and $c_0(X_n)$ contain a copy of c_0 . Must every Banach space with property CWO contain a copy of c_0 ?
- (b) Does there exist a dual (infinite dimensional) Banach space with property CWO?
- (c) If both X and Y have property CWO, does the injective tensor product $X\widehat{\otimes}_\varepsilon Y$ also have property CWO?

We could also ask the same if both X and Y have either property CWO-S or CWO-B.

REFERENCES

This packet records two positive subcases. First, a finite-dimensional space has CWO if and only if it has the source paper's property (co); therefore $C_0(K)\widehat{\otimes}_\varepsilon E$ has CWO whenever K is scattered locally compact and E is finite dimensional with CWO. Second, scalar scattered C_0 -spaces are stable under finite injective tensor products.

Definitions

A Banach space X has property CWO if every finite convex combination of relatively weakly open subsets of B_X is relatively weakly open in B_X . The source paper also defines the pointwise variants CWO-S and CWO-B, and CWO implies both.

A topological space is scattered if every nonempty subspace has an isolated point.

Idea of the proof

The source paper proves that, if M is scattered locally compact Hausdorff and E is finite dimensional with property (co), then $C_0(M, E)$ has property CWO (Theorem 2.5 in the PDF). The new point recorded here is that, in finite dimensions, CWO already forces (co). Given a convex decomposition $x = \sum_i \lambda_i x_i$, apply CWO to small relative balls around the x_i 's. This gives local surjectivity of the convex-combination map. A standard finite-dimensional continuous selection theorem for convex fibers then turns local surjectivity into the continuous local decomposition required by (co).

For scalar scattered C_0 -spaces, the tensor-product question reduces to topology. The injective tensor product $C_0(K) \widehat{\otimes}_\varepsilon C_0(L)$ is canonically $C_0(K \times L)$, and finite products of scattered spaces are scattered.

Result

Theorem 1. *Let E be a finite-dimensional Banach space. Then E has property CWO if and only if E has property (co).*

Proof. The implication (co) $\Rightarrow$ CWO is Proposition 2.2 of the source paper; the proof is short, so we recall the mechanism. In finite dimension, weak and norm topologies agree. If $x = \sum_i \lambda_i x_i$ with $x_i \in U_i$, where the U_i 's are relatively open in B_E , property (co) gives continuous local decompositions of every u near x into $u = \sum_i \lambda_i v_i(u)$, with $v_i(u)$ still inside U_i . Thus $\sum_i \lambda_i U_i$ is relatively open in B_E .

Conversely, assume E has CWO. Fix

$$x = \sum_{i=1}^n \lambda_i x_i, \quad x_i \in B_E, \quad \lambda_i > 0, \quad \sum_i \lambda_i = 1,$$

and fix $\varepsilon > 0$. Choose $0 < \eta < \varepsilon$, and set

$$U_i = \{y \in B_E : \|y - x_i\| < \eta/2\}.$$

By CWO, the set $C = \sum_i \lambda_i U_i$ is relatively open in B_E and contains x . Hence there is $\delta > 0$ such that

$$D := B(x, \delta) \cap B_E \subset C.$$

Now put

$$A_i = \{y \in B_E : \|y - x_i\| \leq \eta\}, \quad A = \prod_{i=1}^n A_i,$$

and let $T : A \rightarrow E$ be the affine restriction of the linear map

$$T(y_1, \dots, y_n) = \sum_{i=1}^n \lambda_i y_i.$$

For every $u \in D$, the fiber

$$\Phi(u) = \{(y_1, \dots, y_n) \in A : T(y_1, \dots, y_n) = u\}$$

is nonempty, compact, and convex. Nonemptiness follows from $D \subset C$ and $U_i \subset A_i$.

The multifunction $\Phi : D \rightarrow 2^A$ is lower semicontinuous. Indeed, if O is relatively open in A , then the elementary open mapping theorem for linear maps on finite-dimensional convex sets gives that $T(O)$ is relatively open in $T(A)$. Consequently

$$\{u \in D : \Phi(u) \cap O \neq \emptyset\} = D \cap T(O)$$

is relatively open in D . A finite-dimensional form of Michael's selection theorem for lower semicontinuous convex-valued maps now gives a continuous selection $s = (s_1, \dots, s_n) : D \rightarrow A$.

For every $u \in D$,

$$u = \sum_i \lambda_i s_i(u), \quad \|s_i(u) - x_i\| \leq \eta < \varepsilon.$$

This is exactly property (co) at the point x . Since the decomposition and ε were arbitrary, E has (co). $\square$

Theorem 2. *Let K be a scattered locally compact Hausdorff space, and let E be a finite-dimensional Banach space with CWO. Then*

$$C_0(K) \widehat{\otimes}_\varepsilon E \cong C_0(K, E)$$

has property CWO.

Proof. By the previous theorem, E has property (co). The conclusion is exactly Theorem 2.5 of Abrahamsen–Becerra Guerrero–Haller–Lima–Poldvere, together with the standard identification $C_0(K) \widehat{\otimes}_\varepsilon E \cong C_0(K, E)$ for finite-dimensional coefficient spaces. $\square$

Theorem 3. *Let K and L be scattered locally compact Hausdorff spaces. Then*

$$C_0(K) \widehat{\otimes}_\varepsilon C_0(L)$$

has property CWO. In particular, the tensor-product question from arXiv:1806.10693 has an affirmative answer for pairs of scalar scattered C_0 -spaces.

More generally, if $K_1, \dots, K_n$ are scattered locally compact Hausdorff spaces, then

$$C_0(K_1) \widehat{\otimes}_\varepsilon \cdots \widehat{\otimes}_\varepsilon C_0(K_n)$$

has property CWO.

Proof. We first note that a finite product of scattered spaces is scattered. It suffices to prove the two-factor case. Let A be a nonempty subset of $K \times L$. The projection $p_K(A)$ is a nonempty subset of K , so there exist $x \in p_K(A)$ and an open set $U \subset K$ such that

$$U \cap p_K(A) = \{x\}.$$

The fiber

$$A_x = \{y \in L : (x, y) \in A\}$$

is a nonempty subset of L , hence has an isolated point y . Choose an open set $V \subset L$ with $V \cap A_x = \{y\}$. Then

$$(U \times V) \cap A = \{(x, y)\},$$

so A has an isolated point. Therefore $K \times L$ is scattered. Since finite products of locally compact Hausdorff spaces are locally compact Hausdorff, $K \times L$ is scattered locally compact Hausdorff.

The canonical bilinear map

$$(f, g) \mapsto [(s, t) \mapsto f(s)g(t)]$$

extends to the standard isometric identification

$$C_0(K) \widehat{\otimes}_\varepsilon C_0(L) \cong C_0(K \times L).$$

By Theorem 2.5 of Abrahamsen–Becerra Guerrero–Haller–Lima–Poldvere, applied to the scalar field as the finite-dimensional coefficient space, $C_0(M)$ has property CWO whenever M is scattered locally compact Hausdorff. Applying this with $M = K \times L$ proves the two-factor case. The finite-factor statement follows by induction, using the same argument and associativity of the injective tensor product for these canonical C_0 -identifications. $\square$

Verification and novelty notes

The scalar C_0 -tensor proof uses three ingredients:

1. Theorem 2.5 of the source paper, specialized to scalar $C_0(M)$.
2. The standard locally compact tensor identity $C_0(K) \widehat{\otimes}_\varepsilon C_0(L) \cong C_0(K \times L)$.
3. The elementary projection-fiber proof that finite products of scattered spaces are scattered.

The finite-dimensional coefficient upgrade additionally uses a standard finite-dimensional form of Michael’s continuous selection theorem for lower-semicontinuous convex-valued maps.

Cheap run indexes were searched for arXiv:1806.10693, “property CWO”, “property (co)”, “finite dimensional”, “injective tensor product”, and related phrases. No prior packet for this finite-dimensional equivalence or the tensor upgrade was found. External exact-phrase searches for the CWO tensor question found the source paper but no later explicit full resolution.

The novelty confidence is low for the scalar C_0 -tensor corollary and moderate for the finite-dimensional $\text{CWO} \Rightarrow (\text{co})$ selection argument. The packet is not a full solution to Question 6(c).

Limitations

This does not address arbitrary pairs of CWO spaces, nor the question for spaces with only CWO-S or only CWO-B. It also does not produce a counterexample or obstruction for the general injective tensor product.

Human-review recommendation

Check the finite-dimensional selection step proving $\text{CWO} \Rightarrow (\text{co})$. The key point is the lower semicontinuity of the convex fiber map. Then check the locally compact C_0 -tensor identifications used in the corollaries.

References

- [1] T. A. Abrahamsen, J. Becerra Guerrero, R. Haller, V. Lima, and M. Poldvere, *Banach spaces where convex combinations of relatively weakly open subsets of the unit ball are relatively weakly open*, arXiv:1806.10693, 2018.
- [2] R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, Springer Monographs in Mathematics, Springer, 2002.

- [3] E. Michael, *Continuous selections. I*, Ann. of Math. (2) 63 (1956), 361–382.